Supplemental Material for "Amplitude-Robust Qubit Spectroscopy Using Echo-Root-Lorentzian Pulse Shapes"

Asaf A. Solonnikov and Nadav Katz

The Racah Institute of Physics, The Hebrew University of Jerusalem, Israel

(Dated: July 20, 2026)

This Supplemental Material provides the experimental setup and qubit coherence characterization; a precise definition of the finite echo-root-Lorentzian pulse and its resonant cancellation mechanism; the Bloch-equation coherence limit; pulse-cutoff and adiabaticity analyses; high-resolution and extended-amplitude scans; the numerical model and data-analysis procedure; and comparisons between pulse protocols, simulation, and experiment.

S1. MEASUREMENT SETUP

Experiments were performed on a QPU comprising 11 uncoupled superconducting qubits hosted in a dilution refrigerator. Control and readout were implemented with a Quantum Machines OPX1000 equipped with a microwave front-end module (MW-FEM), together with external cryogenic attenuators, filters, circulators, and amplifiers.

A schematic of the setup is shown in Fig. S1. The MW-FEM uses direct digital synthesis and digital upconversion: a programmed complex pulse envelope is translated to the configured microwave carrier and emitted directly from an RF output, without an external analog local oscillator or IQ mixer [1]. For acquisition, the returned microwave signal is digitally downconverted to a complex baseband stream. QUA demodulation then applies cosine and sine integration weights to the sampled signal and accumulates the result to obtain the measured quadratures [2]. Thus, separate MW-FEM outputs provide the qubit-drive and resonator-readout tones, while an MW-FEM input acquires the returning readout signal. A dedicated DC bias line was available but was not used in these measurements; consequently, the qubits were not necessarily operated at their flux sweet spots.

The microwave signals are routed into the dilution refrigerator through heavily attenuated coaxial lines. Attenuators are distributed across multiple temperature stages (300 K to $\sim$9 mK) to thermalize the lines and suppress room-temperature noise. Low-pass filters on the DC-bias line further suppress high-frequency noise before the signals reach the device.

The quantum device is mounted at the mixing-chamber stage ($\sim$9 mK). The outgoing readout signal is routed through a chain of circulators that isolates the qubit from amplifier back-action, followed by cryogenic amplification using a high-electron-mobility-transistor (HEMT) amplifier at the 4 K stage. The amplified signal returns directly to an MW-FEM RF input for digital downconversion, demodulation, and real-time processing.

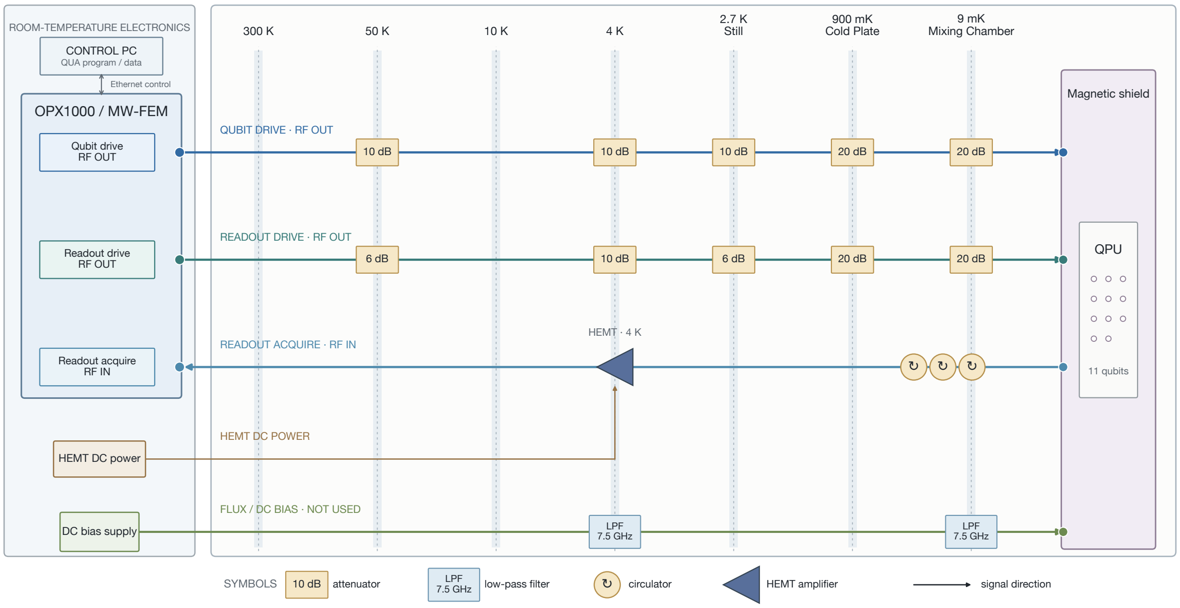

Figure S1. Schematic of the experimental setup for qubit control and readout. A control computer submits QUA programs and receives data over Ethernet. The OPX1000 MW-FEM directly generates the qubit-drive and resonator-readout RF signals and acquires the returning readout signal. Input lines are attenuated across the 300 K, 50 K, 10 K, 4 K, still, cold-plate, and mixing-chamber stages. The low-pass-filtered DC-bias line is shown for completeness but was not used. At the mixing chamber, the device is isolated by circulators; the returning readout signal is amplified by a 4 K HEMT before entering the MW-FEM for digital downconversion and demodulation.

S1.1. Qubit coherence characterization

The energy-relaxation and Ramsey-dephasing times of the qubit used for the spectroscopy measurements were characterized independently. Figure S2 shows representative measurements for qubit q1. Fitting the excited-state population to an exponential decay with a constant offset gives $T_1 = 51.24 \pm 0.73$ μs. Fitting the positive-detuning Ramsey trace to an exponentially damped sinusoid gives $T_2^* = 7.31 \pm 0.13$ μs. The quoted uncertainties are one-standard-deviation estimates obtained from the fit covariance matrices. No associated Hahn-echo trace is available, so no measured Hahn-echo T_2 is reported. When one transverse time is required, we conservatively use $T_{2,\mathrm{eff}} = T_2^* = 7.31$ μs. With the measured T_1, this gives $T_\phi = 7.87$ μs through $T_{2,\mathrm{eff}}^{-1} = (2T_1)^{-1} + T_\phi^{-1}$; this is not Hahn-echo evidence.

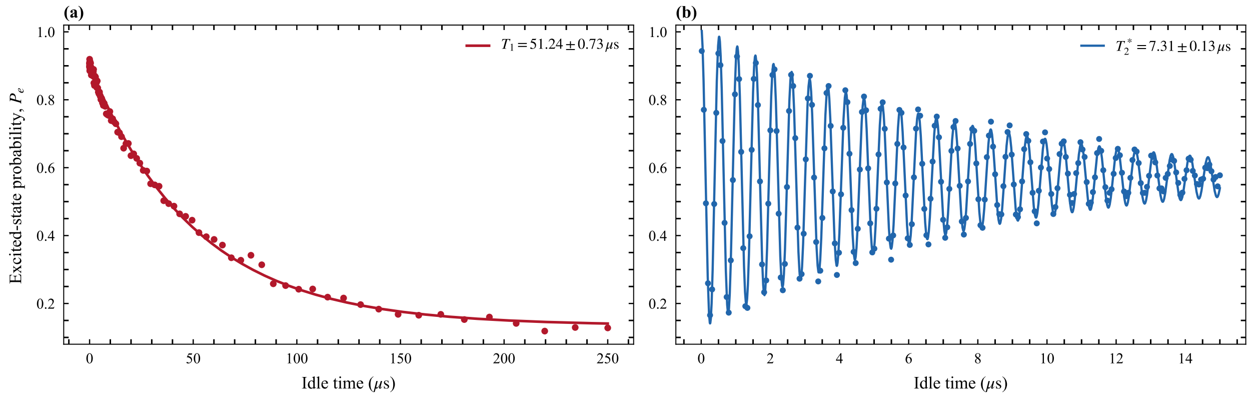

Figure S2. Independent coherence characterization of qubit q1. (a) Energy relaxation following state preparation, together with an exponential fit. (b) Ramsey fringes acquired with a positive 2 MHz programmed detuning, together with an exponentially damped sinusoidal fit.

S2. FINITE PULSE DEFINITION AND ECHO CANCELLATION

We define the finite generalized Lorentzian envelope by

$$f_{n,c,L}(t) = \begin{cases} \left[1 + (t/\tau)^2\right]^{-n}, & |t| \leq L/2, \\ 0, & |t| > L/2, \end{cases} \tag{S1}$$

where L is the total pulse duration, n is the shape exponent, and the relative endpoint amplitude is $c = f_{n,c,L}(L/2)$. These parameters imply

$$\tau = \frac{L/2}{\sqrt{c^{-1/n} - 1}}. \tag{S2}$$

The root-Lorentzian pulse used here corresponds to $n = 1/2$. Its echo version has a phase inversion at the center,

$$\Omega_{\mathrm{E}}(t) = \Omega_0 f_{n,c,L}(t) s(t), \qquad s(t) = \begin{cases} +1, & t < 0, \\ -1, & t \geq 0. \end{cases} \tag{S3}$$

The value assigned at the single point $t = 0$ has no effect in the continuum limit. In a sampled waveform, the two halves are constructed with equal duration and opposite phase.

Two cutoff scans must be distinguished. A pure truncation scan keeps τ, n, and Ω_0 fixed while changing L. A fixed-duration hardware scan instead keeps L fixed and changes the endpoint ratio c; in the parametrization of Eq. (S2), this also changes τ. Results from these two scans need not coincide.

In the rotating-wave approximation, the unitary two-level Hamiltonian is

$$\frac{H(t)}{\hbar} = \frac{1}{2}\left[\Delta\sigma_z + \Omega_{\mathrm{E}}(t)\sigma_x\right]. \tag{S4}$$

At exact resonance, Hamiltonians at different times are proportional to the same operator and therefore commute. The propagator is

$$U(\Delta = 0) = \exp\left[-\frac{i\sigma_x}{2}\int_{-L/2}^{L/2}\Omega_{\mathrm{E}}(t)\,dt\right] = \mathbb{I}, \tag{S5}$$

because the signed areas of the two symmetric halves cancel. For $\Delta \neq 0$, the σ_z and σ_x terms do not commute, so the cancellation is incomplete and a narrow detuning-dependent response remains. Equation (S5) is exact only for a symmetric, unitary two-level pulse. Relaxation, dephasing, phase or amplitude imbalance, and finite waveform bandwidth prevent exact cancellation in the experiment.

S2.1. Numerical illustration of the echo mechanism

Figure S3 shows the numerical qubit-state evolution for the root-Lorentzian pulse and its echo variant. At resonance, the drive produces coherent evolution during the first half of the echo pulse. The π phase inversion reverses this evolution during the second half, so the ideal symmetric sequence returns the qubit to its initial state. In the open-system simulation the return is incomplete because relaxation and dephasing act throughout the pulse, consistent with the limitations of Eq. (S5).

For nonzero detuning, the $\Delta\sigma_z$ term does not reverse at the pulse midpoint and does not commute with the driven evolution. The two pulse halves therefore fail to cancel, leaving a detuning-dependent final population. The contrast between the resonant return and the off-resonant response produces the central depletion feature used to identify the qubit transition.

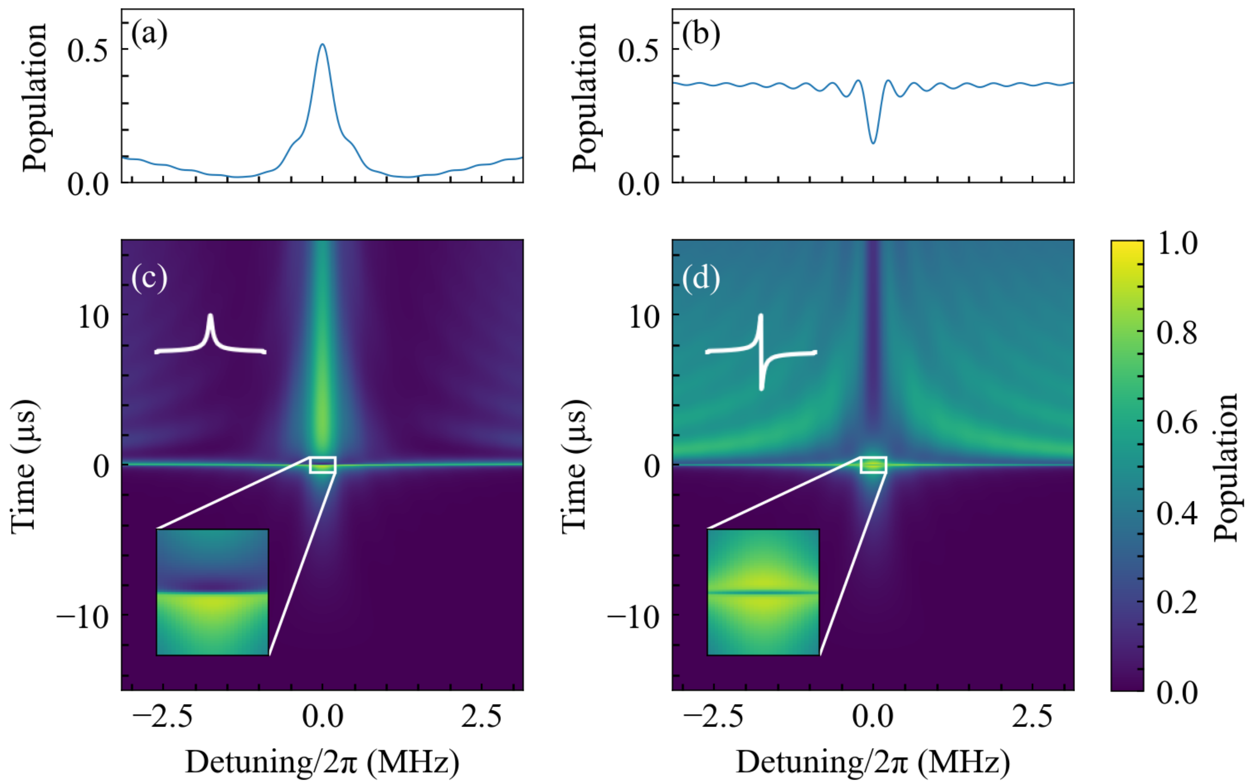

Figure S3. Numerical illustration of the echo-cancellation mechanism. (a) Final excited-state population versus detuning $\Delta/2\pi$ for a root-Lorentzian pulse. (b) Corresponding response for the echo-root-Lorentzian pulse $L_{\text{echo}}^{(1/2)}$. (c,d) Full excited-state evolution versus time and detuning for the two protocols. The pulse duration is 30 μs and the relative endpoint amplitude is 5×10^{-4}. The ordinary root-Lorentzian pulse produces a resonant peak, whereas the phase inversion in the echo pulse reverses the resonant evolution and produces a central dip. At nonzero detuning the two halves do not cancel, leaving the off-resonant population that forms the spectroscopic response. These archived arrays use the historical coherence parameters documented in Sec. S7, rather than the later measurement-linked values.

S3. BLOCH-EQUATION COHERENCE LIMIT

For comparison with the shaped-pulse protocol, we first summarize the steady-state response of a continuously driven two-level system. We use the Bloch-vector convention $w = \rho_{00} - \rho_{11}$, so that the excited-state population is $P_e = (1 - w)/2$. Solving the optical Bloch equations for a drive with angular Rabi frequency Ω_0, angular detuning Δ, relaxation time T_1, and transverse coherence time T_2 gives [3]

$$w_{\mathrm{ss}}(\Delta) = \frac{1+\Delta^2 T_2^2}{1+\Delta^2 T_2^2 + \Omega_0^2 T_1 T_2}, \qquad P_{e,\mathrm{ss}} = \frac{1-w_{\mathrm{ss}}}{2}. \tag{S6}$$

This expression assumes a pulse long enough to approach steady state. It is a reference result for conventional saturation spectroscopy, rather than a closed-form description of the finite echo pulse.

The full width at half maximum of $P_{e,\mathrm{ss}}(\Delta)$ is

$$\Gamma_{\mathrm{FWHM}} = \frac{2}{T_2}\sqrt{1+\Omega_0^2 T_1 T_2}. \tag{S7}$$

All frequencies in Eqs. (S6) and (S7) are angular frequencies. The corresponding linewidth in hertz is $\Gamma_{\mathrm{FWHM}}/(2\pi)$.

It is useful to introduce the saturation parameter $s = \Omega_0^2 T_1 T_2$. In the weak-drive limit $s \ll 1$,

$$\Gamma_{\mathrm{FWHM}} \longrightarrow \frac{2}{T_2}, \tag{S8}$$

which is the coherence-limited angular-frequency linewidth. In the strong-drive limit $s \gg 1$,

$$\Gamma_{\mathrm{FWHM}} \longrightarrow 2\Omega_0\sqrt{\frac{T_1}{T_2}}, \tag{S9}$$

showing the familiar linear power broadening. These limits are governed by s, not by comparing Ω_0 with a detuning-dependent quantity.

S4. PULSE TRUNCATION AND ENDPOINT AMPLITUDE

The ideal Lorentzian tail has infinite support, whereas the implemented pulse starts and ends at finite amplitude $\Omega_c = c\Omega_0$. Abruptly switching this residual drive produces an endpoint transient. A useful estimate of the associated population contribution is [4]

$$P_c(\Delta) \simeq \frac{\Omega_c^2}{\Omega_c^2+\Delta^2}\left[1-P_0(\Delta)\right], \qquad \Omega_c = c\Omega_0, \tag{S10}$$

where P_0 denotes the response that would be obtained without the endpoint jump. The artifact becomes important when Ω_c is comparable to the linewidth scale of interest. Thus the cutoff cannot be assessed from c alone: it must be considered together with the peak drive and the desired resolution.

We therefore measured the cutoff dependence over two complementary q1 echo–root–Lorentzian campaigns. The broad survey spans $c = 0.002$–0.99 with 25 peak-amplitude settings, while the focused sweep resolves $c = 0.01$–0.10 with 200 peak-amplitude settings. Both use a 20 μs pulse and the same measured $T_2 = 6.856$ μs. Figure S4 reports the dimensionless reciprocal-linewidth metric

$$\mathcal{R} = \frac{1/(\pi T_2)}{\mathrm{FWHM}}, \tag{S11}$$

for which $\mathcal{R} = 1$ is the T_2-limited linewidth and $\mathcal{R} < 1$ denotes a broader feature. We also show the signal-weighted metric $|A_{\mathrm{fit}}|\mathcal{R}$, which suppresses narrow fits with negligible contrast. These are linewidth-based scores, not direct estimates of frequency-estimation uncertainty.

The maps retain only resolved and bounded probability features: the fitted contrast is between 0.05 and 1, $R^2 \geq 0.60$, the fitted center lies inside the measured frequency interval, and the FWHM lies between two frequency bins and half the sweep span. Gray cells denote measurements for which the fit did not pass this common screen. The denser focused sweep provides substantially more accepted fits than the coarse survey and resolves the loss of reciprocal linewidth performance as the cutoff increases across the 0.01–0.10 interval.

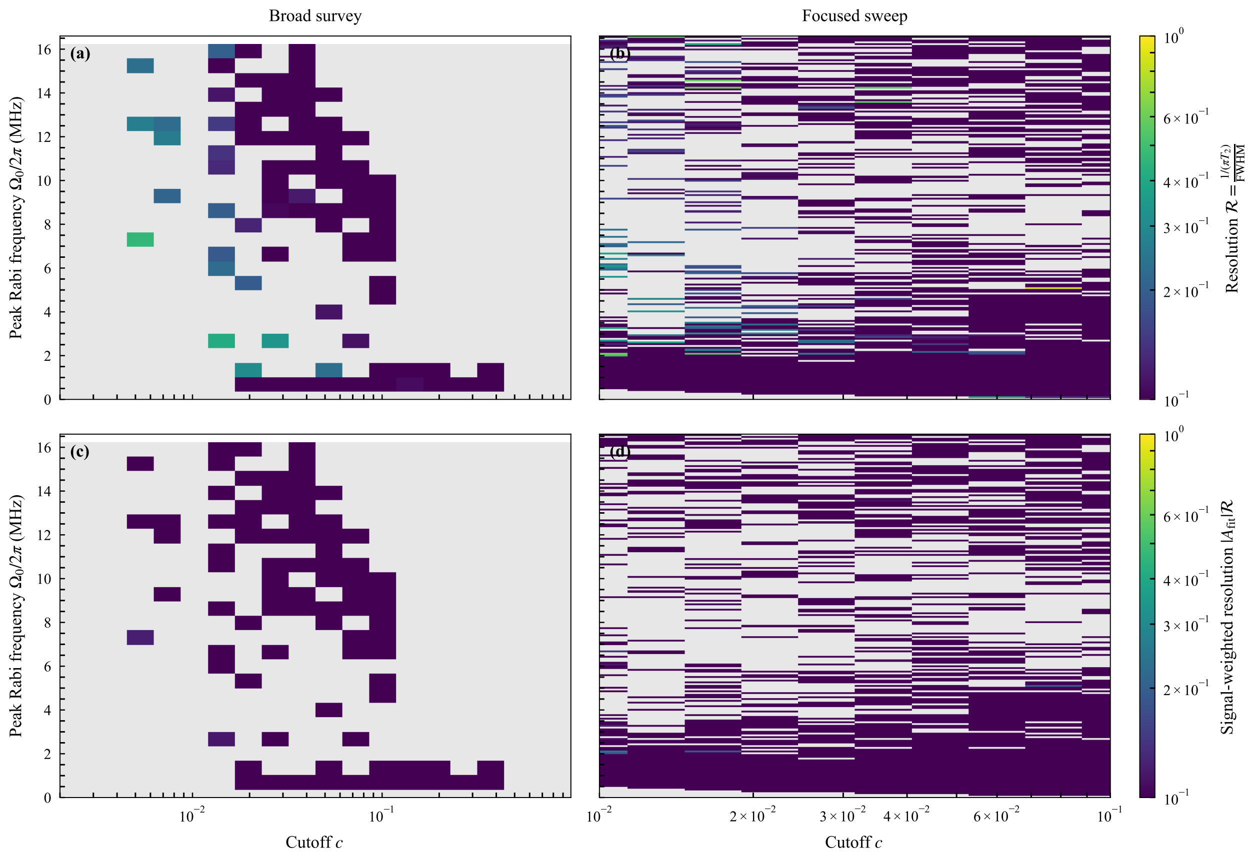

Figure S4. Experimental cutoff sweep for q1 echo–root–Lorentzian spectroscopy. The left column is the broad `cutoff_amp_fwhm_map` campaign (c = 0.002–0.99, 25 amplitudes, 60 repetitions per point); the right column is the denser `echo_lorentzian_cutoff_sweep` campaign (c = 0.01–0.10, 200 amplitudes, 100 repetitions per point). (a,b) Reciprocal-linewidth metric $\mathcal{R}$ and (c,d) signal-weighted metric $|A_{\mathrm{fit}}|\mathcal{R}$. Both color scales are logarithmic. On the upper color scale, $\mathcal{R} = 1$ is the T_2-limited linewidth and values below one correspond to FWHM broader than that limit. Gray cells are rejected or unresolved fits under the common screen described in the text; no fitted surface or interpolation is shown.

S5. ADIABATIC-BASIS DESCRIPTION

For one half of the pulse, the rotating-frame Hamiltonian can be written

$$\frac{H(t)}{\hbar} = \frac{1}{2}\begin{bmatrix} -\Delta & \Omega(t) \\ \Omega(t) & \Delta \end{bmatrix}. \tag{S12}$$

Define

$$\epsilon(t) = \sqrt{\Delta^2 + \Omega^2(t)}, \qquad \vartheta(t) = \frac{1}{2}\,\mathrm{atan2}[\Omega(t), \Delta]. \tag{S13}$$

The instantaneous eigenvalues are $\pm\hbar\epsilon/2$. In the corresponding adiabatic basis, with phases chosen consistently across the pulse,

$$\frac{H_a(t)}{\hbar} = \begin{bmatrix} -\epsilon/2 & -i\dot{\vartheta} \\ i\dot{\vartheta} & \epsilon/2 \end{bmatrix}, \qquad \dot{\vartheta} = \frac{\Delta\dot{\Omega}}{2(\Delta^2+\Omega^2)}. \tag{S14}$$

Adiabatic following requires [5]

$$2|\dot{\vartheta}| \ll \epsilon. \tag{S15}$$

Replacing the inequality by equality provides the local crossover estimate

$$\sqrt{\Delta_b^2+\Omega^2(t_0)} = \frac{|\Delta_b\dot{\Omega}(t_0)|}{\Delta_b^2+\Omega^2(t_0)}, \tag{S16}$$

where t_0 is the time at which the nonadiabatic coupling is largest for the detuning under consideration. This is a scaling criterion, not an exact linewidth formula.

For an untruncated generalized Lorentzian tail with exponent $n > 1/2$, the usual adiabatic estimate gives

$$\Delta_b\tau \propto (\Omega_0\tau)^{-1/(2n-1)}. \tag{S17}$$

The root-Lorentzian case $n = 1/2$ is marginal and lies outside the domain of Eq. (S17); substituting $n = 1/2$ into that power law is therefore invalid. The linewidth of the finite, echo-modulated root-Lorentzian pulse is evaluated numerically and experimentally in this work.

S6. LARGE-DRIVE SPECTROSCOPY

S6.1. Broad and Narrow Frequency Scans

High-resolution spectroscopy at large drive amplitudes is characterized on two frequency scales. A broad scan first locates the qubit transition and reveals the surrounding baseline, nearby spectral features, and any strong-drive distortions. A narrow scan centered on the transition then samples the sharp echo–root–Lorentzian depletion feature densely enough to determine its center and linewidth. Presenting the broad and narrow scans together distinguishes a genuinely narrow resonance from a local feature embedded in a wider distorted response and demonstrates that the high resolution persists within the full spectral landscape.

S6.2. Extended-Amplitude Regime and Strong-Drive Distortions

At drive amplitudes above approximately 50 MHz, the Rabi frequency is no longer small compared with the transmon anharmonicity and the measured response begins to depart from an effective two-level description. Possible mechanisms include coupling to higher transmon levels, drive-induced level shifts, leakage, and multiphoton processes. The two-level calculation in Sec. S7 cannot distinguish among these mechanisms, so the additional structure is reported here as a strong-drive distortion rather than assigned to a specific transition. Quantitative attribution would require a converged multilevel model and an independent calibration of the delivered drive amplitude.

S6.3. Cutoff Dependence from 50 to 80 MHz

To separate large-drive behavior from changes in pulse duration or sampling, we compare three q1 echo–root–Lorentzian scans acquired with the same 10 μs duration, detuning grid, amplitude grid, active-reset protocol, and 40 repetitions per point. Only the dimensionless cutoff-control setting is changed, from $c = 0.001$ to 0.005 and 0.010. Using the same-day q1 π-pulse calibration retained with the data, the displayed portion of each scan is selected from raw peak Rabi frequencies of 50.28 to 76.97 MHz. The centers of the displayed block averages span 50.67 to 76.19 MHz. This restricted view exposes the high-amplitude response directly instead of allowing the much larger low-amplitude region to dominate the color scale. To reduce binomial sampling noise without imposing a fit or interpolated surface, each displayed cell is a non-overlapping average of three adjacent detuning points and three adjacent amplitude points. Since every raw point contains 40 repetitions, a displayed cell averages 360 binary outcomes.

Figure S5 shows a pronounced reorganization of the large-drive response with cutoff. At $c = 0.001$ the map is dominated by a step-like change across zero detuning, whereas the $c = 0.005$ and 0.010 scans develop curved, amplitude-dependent fringes that cross the central-frequency region. Thus these measurements do not support extrapolating the low-amplitude narrow-feature behavior unchanged into the 50–80 MHz regime. Instead, they identify a cutoff-sensitive strong-drive regime in which a single-linewidth description is insufficient. Because all panels use one color scale, the differences are not artifacts of independent

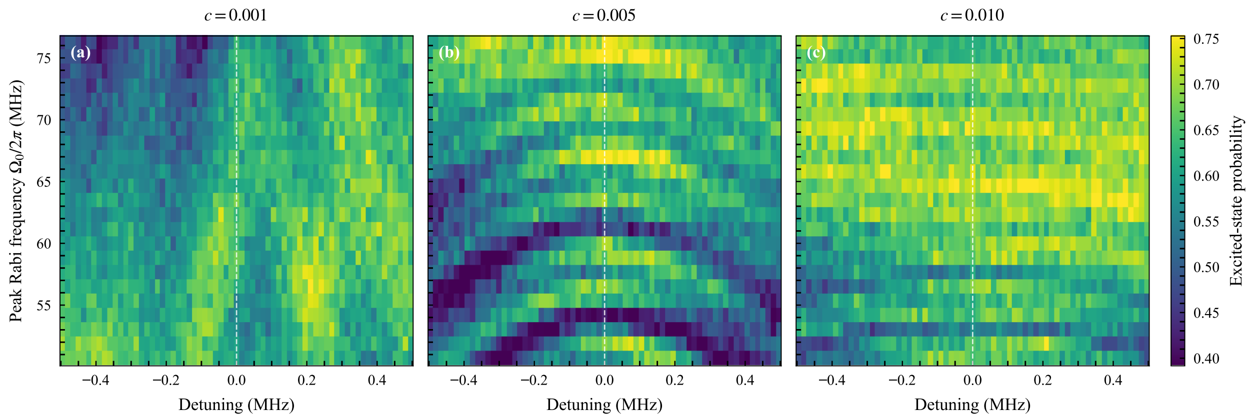

Figure S5. Measured high-amplitude echo–root–Lorentzian spectroscopy on q1 for dimensionless cutoff-control settings (a) $c = 0.001$, (b) $c = 0.005$, and (c) $c = 0.010$. Each panel shows the same 50.28–76.97 MHz raw peak-Rabi-frequency interval from a 10 μs scan with 40 repetitions per point. The dashed line marks zero detuning, and all panels share the color scale. Displayed cells are non-overlapping 3×3 block averages of the measured grid; no interpolation or fitted surface is shown. The progression from a step-like response in (a) to curved interference fringes in (b) and (c) demonstrates that the large-drive structure is cutoff sensitive.

normalization.

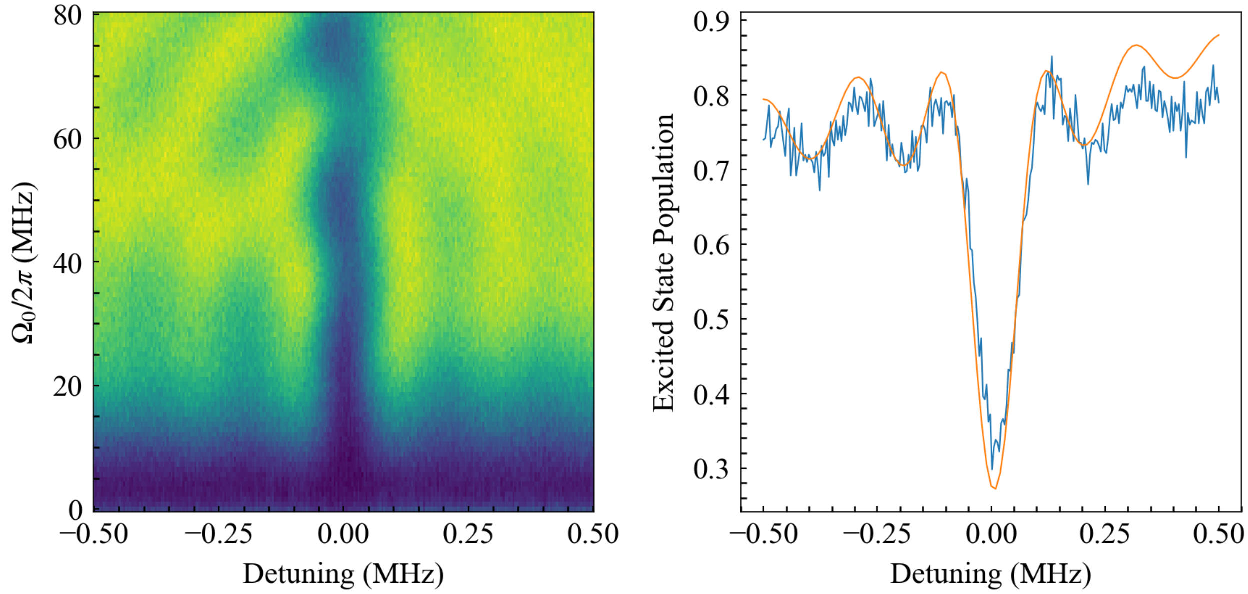

Figure S6. Two-dimensional spectroscopy map obtained with a 10 μs echo–root–Lorentzian pulse at extended drive amplitudes (up to 80 MHz). The left panel shows the full detuning–amplitude map, while the right panel presents a line cut at fixed drive strength (blue) together with a smooth fit (orange). The observed distortions mark the breakdown of the ideal effective two-level response as the drive approaches the multilevel regime.

S7. NUMERICAL MODEL

The simulations use an effective driven two-level system coupled to Markovian relaxation and dephasing channels. In the frame rotating at the drive frequency, and within the rotating-wave approximation, the Hamiltonian is implemented as

$$\frac{H(t)}{\hbar} = \Delta a^\dagger a + \frac{\Omega_0}{2} f(t) \left(a + a^\dagger\right), \tag{S18}$$

where $\Delta = \omega_q - \omega_d$ is the qubit–drive detuning, Ω_0 is the peak Rabi frequency, and $a = |0\rangle\langle 1|$ is the lowering operator. Up to an irrelevant identity term and the chosen sign convention for Δ, Eq. (S18) is equivalent to $H/\hbar = (\Delta/2)\sigma_z + (\Omega_0 f/2)\sigma_x$.

The simulations use the finite pulse in Eqs. (S1)–(S3). The overall sign of the echo envelope is a phase convention and does not affect the measured populations.

The density matrix evolves according to the Lindblad master equation

$$\dot{\rho} = -\frac{i}{\hbar}[H(t), \rho] + \mathcal{D}\left[\sqrt{\frac{1}{T_1}}\, a\right] \rho + \mathcal{D}\left[\sqrt{\frac{2}{T_\phi}}\, a^\dagger a\right] \rho, \tag{S19}$$

where $\mathcal{D}[C]\rho = C\rho C^\dagger - \frac{1}{2}\{C^\dagger C, \rho\}$ and

$$\frac{1}{T_2} = \frac{1}{2T_1} + \frac{1}{T_\phi}. \tag{S20}$$

The coherence characterization used by measurement-linked calculations is given in the measurement-setup section. The archived figure arrays were generated with the historical inputs $T_1 = 30$ μs and $T_\phi = 20$ μs, which imply $T_2 = 12$ μs. These values describe the cached numerical data, not the later q1 coherence measurement. A measurement-linked calculation instead uses $T_1 = 51.24 \pm 0.73$ μs and, in the absence of a Hahn-echo measurement, the conservative choice $T_{2,\text{eff}} = T_2^* = 7.31$ μs, corresponding to $T_\phi = 7.87$ μs. A measured Hahn-echo T_2 should replace this effective choice when such a trace is acquired. Each calculation starts in $|0\rangle$, integrates Eq. (S19) over the complete pulse, and records the final Bloch-vector components. Spectroscopy maps are generated by repeating this evolution over the experimental detuning and drive-amplitude grids, using the pulse duration, cutoff, T_1, and T_2 associated with the corresponding data set. The time-dependent master equation is solved with QuTiP or, for the dense matched sweep in Figs. S7–S9, the algebraically equivalent two-level Bloch equations with fourth-order Runge–Kutta integration. Production calculations use a two-dimensional Hilbert space and a uniform temporal sampling of the complete waveform. The default solver grid contains 1000 points; the matched linewidth sweep uses 2000 steps. The even grids contain equal numbers of negative- and positive-time samples, with the central phase inversion between them.

The numerical Hilbert space is restricted to two levels. Consequently, this model captures coherent drive dynamics, relaxation, and pure dephasing, but it does not describe leakage, AC-Stark shifts produced by higher transmon levels, or multiphoton transitions. Comparisons in the extended-amplitude regime are therefore interpreted as tests of where the effective two-level description breaks down rather than as a simulation of those multilevel effects.

S8. EXPERIMENTAL–SIMULATION COMPARISON

The amplitude-robustness data set was acquired on qubit q1 using active reset and state-discriminated readout. Each point contains 40 repetitions. The detuning axis spans -0.5 to $+0.5$ MHz in 1 kHz steps, and the data comprise 100 fixed-amplitude slices. The pulse record contains a root–Lorentzian edge-to-peak ratio $c = 0.005$, a pulse length $L = 10\ \mu$s, and an unscaled peak output amplitude of 0.2. The peak Rabi frequency is reconstructed from the saved q1 square-pulse calibration, including this peak output amplitude; the resulting experimental range is 0 to 15.32 MHz.

For the matched comparisons in Figs. S7–S9, we integrate the two-level Lindblad model of Eq. (S19) on the complete experimental detuning and calibrated peak-Rabi grids. The calculation uses the recorded echo–root–Lorentzian pulse parameters $L = 10\ \mu$s and $c = 0.005$ together with the measurement-linked values $T_1 = 51.24 \pm 0.73\ \mu$s and $T_\phi = 7.87\ \mu$s. No affine readout calibration or fitted frequency offset is applied.

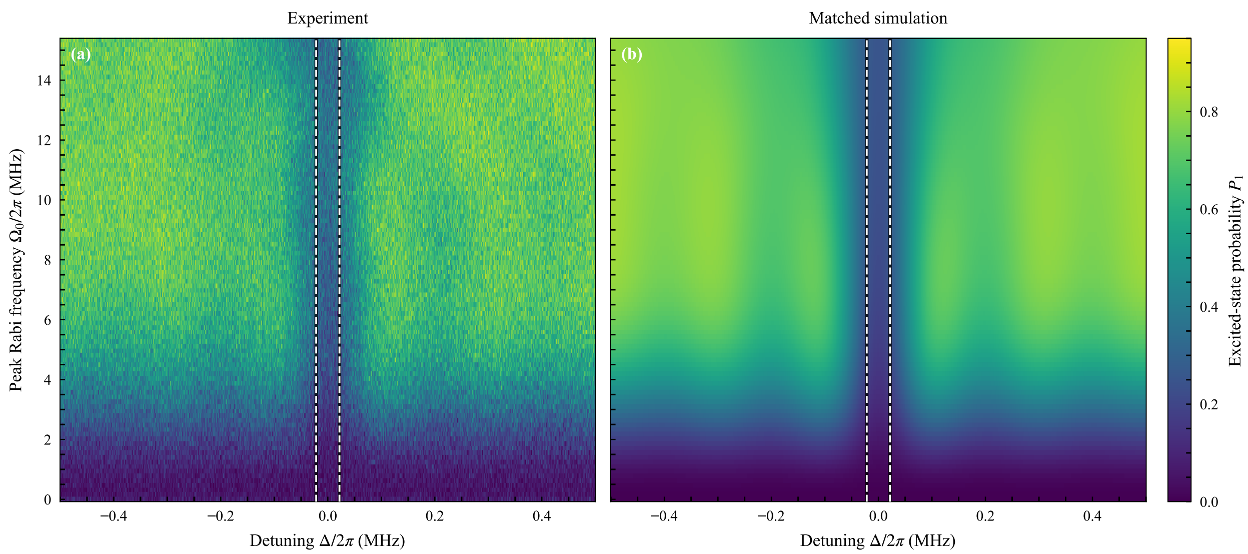

Figure S7. Complete amplitude–detuning sweeps for the q1 echo–root–Lorentzian sequence. (a) Measured excited-state probability from 40 repetitions per point. (b) Parameter-matched two-level Lindblad simulation on the identical calibrated peak-Rabi and detuning grids. The two panels use the same probability color scale. Both show the narrow resonance-centered cancellation channel, while the deterministic model is smoother and predicts a higher off-resonant excitation probability at intermediate and high drive. The paired black-edged white dashed lines at $\Delta/2\pi = \pm\Gamma_{T_2}/2 = \pm 21.75$ kHz enclose the conservative T_2-limited FWHM, $\Gamma_{T_2} = 43.5$ kHz.

Figure S8 resolves this comparison at three representative amplitudes spanning the accepted operating range. The central dip and its drive-dependent broadening are reproduced by the model. The remaining disagreement is primarily in the off-resonant background and side-lobe structure, and therefore appears most directly in the extracted contrast rather than in the central linewidth.

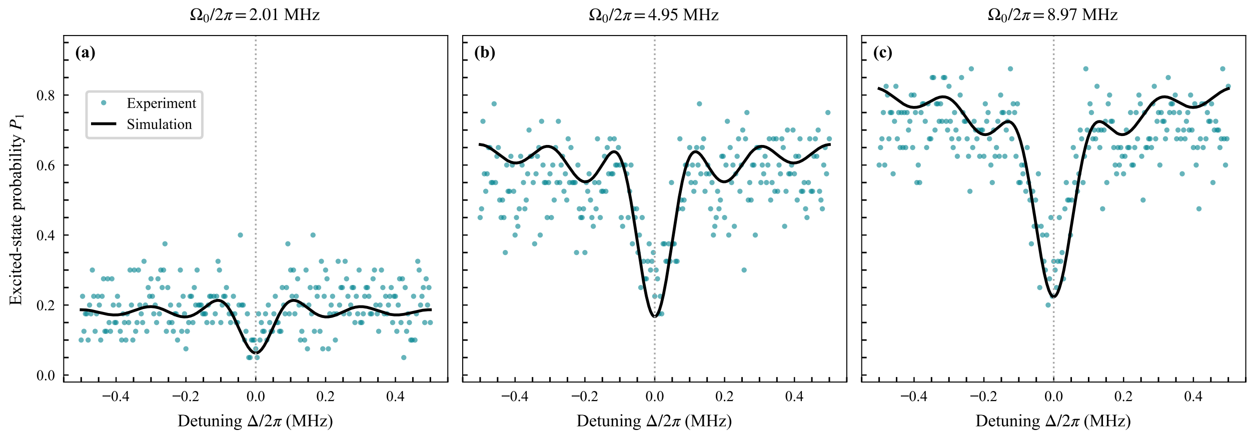

Figure S8. Measured (colored markers) and simulated (black curves) fixed-amplitude spectra at calibrated peak Rabi frequencies $\Omega_0/2\pi = 2.011$, 4.951, and 8.973 MHz. The vertical dotted line denotes zero detuning. For visual clarity, every fourth measured detuning point is displayed; all 1001 points enter the linewidth analysis. The simulation uses the parameters of Fig. S7 without a readout rescaling or fitted frequency shift.

To quantify the comparison across the complete sweep, each fixed-amplitude trace is analyzed independently. We discard 40 points from each frequency edge and apply a Gaussian filter with a standard deviation of one sample to suppress single-bin noise. The remaining data are fit to the operational dip estimator

$$y(\Delta) = y_0 - A \exp\left[-\frac{(\Delta - \Delta_0)^2}{2\sigma^2}\right], \tag{S21}$$

where A is the contrast, Δ_0 is the fitted center, and

$$\Gamma_{\mathrm{FWHM}} = 2\sqrt{2\ln 2}\,|\sigma|. \tag{S22}$$

The Gaussian is an operational, consistent linewidth estimator; it is not a claim that the complete physical response is exactly Gaussian.

We normalize the fitted linewidth to the conservative measurement-linked coherence reference

$$\Gamma_{T_2} = \frac{1}{\pi T_{2,\mathrm{eff}}} = 43.5 \text{ kHz}, \qquad T_{2,\mathrm{eff}} = 7.31\ \mu\text{s}. \tag{S23}$$

The corresponding dimensionless linewidth, normalized resolution, and contrast-weighted resolution are

$$W = \frac{\Gamma_{\mathrm{FWHM}}}{\Gamma_{T_2}}, \qquad R = \frac{\Gamma_{T_2}}{\Gamma_{\mathrm{FWHM}}}, \qquad Q = AR. \tag{S24}$$

Thus, $W = 1$ and $R = 1$ denote the coherence-reference linewidth and its reciprocal resolution, respectively. The metric Q additionally penalizes a narrow feature when its measured dip contrast is small.

The simulated traces are smoothed, fit, normalized, and quality-gated by exactly the same procedure as the experimental traces. Because this is a deterministic model curve, its local least-squares uncertainties are not displayed as sampling-error bars.

Fits are accepted only when all parameters and covariance-derived uncertainties are finite, $A \geq 0.02$, $\Gamma_{\mathrm{FWHM}} \leq 0.12$ MHz, and $|\Delta_0| \leq 0.050$ MHz. The reported parameter uncertainties are one standard deviation from the fit covariance matrix. Of the 100 amplitude slices, 62 pass these numerical gates. Thirty are rejected as too broad, seven have centers outside the accepted window, and one fails both the contrast and center gates. Rejected or failed fits remain missing values; they are never replaced by zero linewidth or zero uncertainty. Of the 100 matched simulation slices, 54 pass the same gates; 41 are too broad, four have contrast below threshold, and the zero-drive slice fails both the contrast and center gates.

For direct visual comparison with the constant-drive reference below, we use the same operational threshold $Q \geq 0.10$. On the sampled matched-simulation grid this selects $\Omega_0/2\pi = 2.475$–8.973 MHz; the maximum simulated value, $Q = 0.219$, occurs at 6.034 MHz.

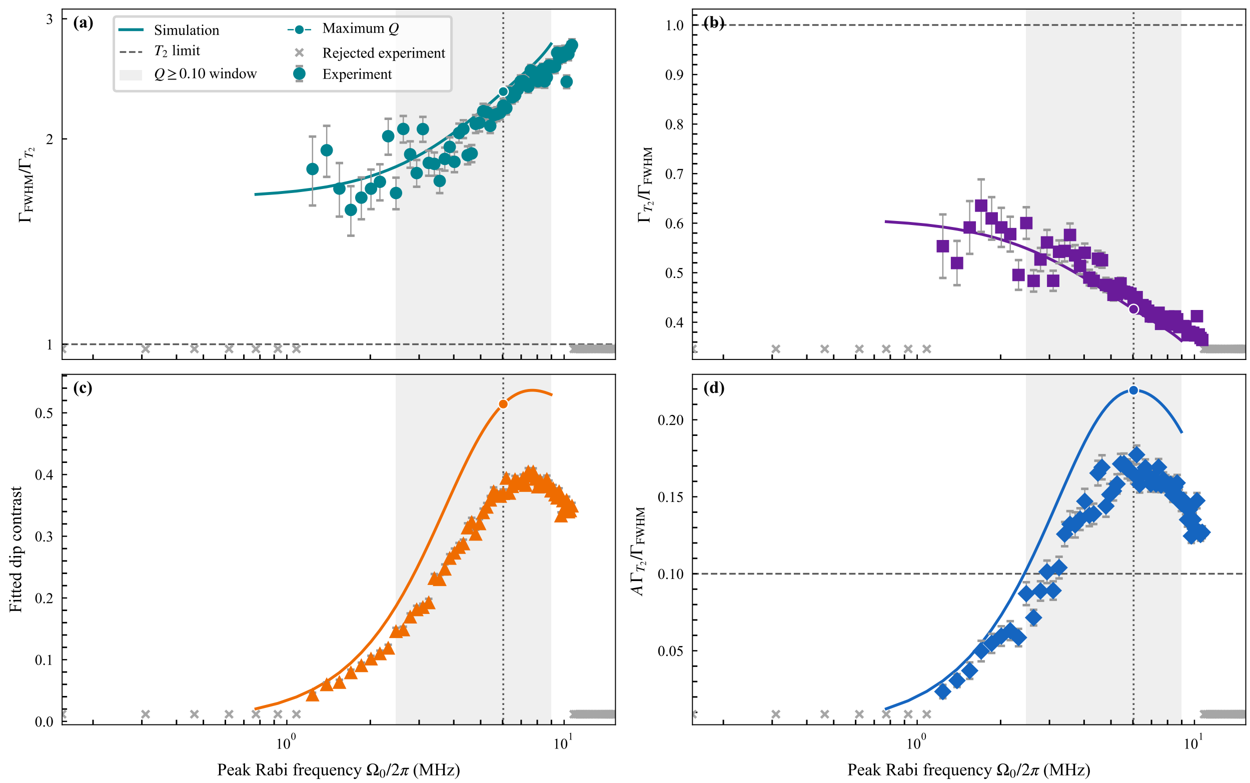

Figure S9. Amplitude dependence of the measured q1 echo–root–Lorentzian response (colored markers) and the parameter-matched two-level Lindblad simulation (same-colored curves). The experimental markers are circles, squares, triangles, and diamonds in panels (a)–(d), respectively. (a) Fitted FWHM in units of the conservative coherence reference Γ_{T_2}. (b) Normalized resolution $R = \Gamma_{T_2}/\Gamma_{\rm FWHM}$. (c) Fitted dip contrast A. (d) Contrast-weighted resolution $Q = AR$. Experimental vertical error bars are propagated from the fit covariance matrix; panel (d) includes the fitted contrast–width covariance. Gray crosses along the lower edge mark experimental amplitude slices rejected by the stated quality gates and are not plotted as zero-valued measurements. Simulation slices that fail the same gates appear as gaps in the corresponding colored curves. As in Fig. S10, the gray band marks the matched-simulation interval with $Q \geq 0.10$; the dotted vertical line and circular curve markers identify the maximum simulated Q. The dashed horizontal line in panel (d) is the same $Q = 0.10$ operational threshold.

S8.1. Constant-pulse reference window

For comparison, we use the analytic steady-state Torrey constant-drive tool employed by the IPS FWHM plots. With saturation parameter $s = \Omega_0^2 T_1 T_2$, the power-broadened linewidth and population contrast are

$$W \equiv \frac{\Gamma_{\mathrm{FWHM}}}{\Gamma_{T_2}} = \sqrt{1+s}, \qquad A = \frac{s}{2(1+s)}. \tag{S25}$$

Here $T_1 = 51.24 \pm 0.73$ μs and $T_2 = 7.31$ μs, consistent with the normalization used above. We also report the normalized resolution $R = 1/W$ and the contrast-weighted resolution $Q = AR$ over the displayed constant-drive range $\Omega_0/2\pi = 10^{-3}$–10^{0} MHz. A logarithmic amplitude axis resolves the low-drive trade-off while retaining the power-broadened regime.

Figure S10 exposes the expected trade-off: contrast rises toward its saturation value while power broadening reduces resolution. In the low-drive limit, $W \to 1$, so the FWHM approaches the T_2-limited value $\Gamma_{T_2} = 1/(\pi T_2) = 43.5$ kHz. Using $Q \geq 0.10$ as an operational visualization threshold gives one interval, $\Omega_0/2\pi = 0.0045$–0.0385 MHz. The analytic maximum occurs at $\Omega_0/2\pi = 0.0116$ MHz, where $W = 1.732$, $R = 0.577$, $A = 0.333$, and $Q = 0.192$. The threshold is used only to highlight the small amplitude window that preserves both signal and resolution; it is not a universal physical boundary.

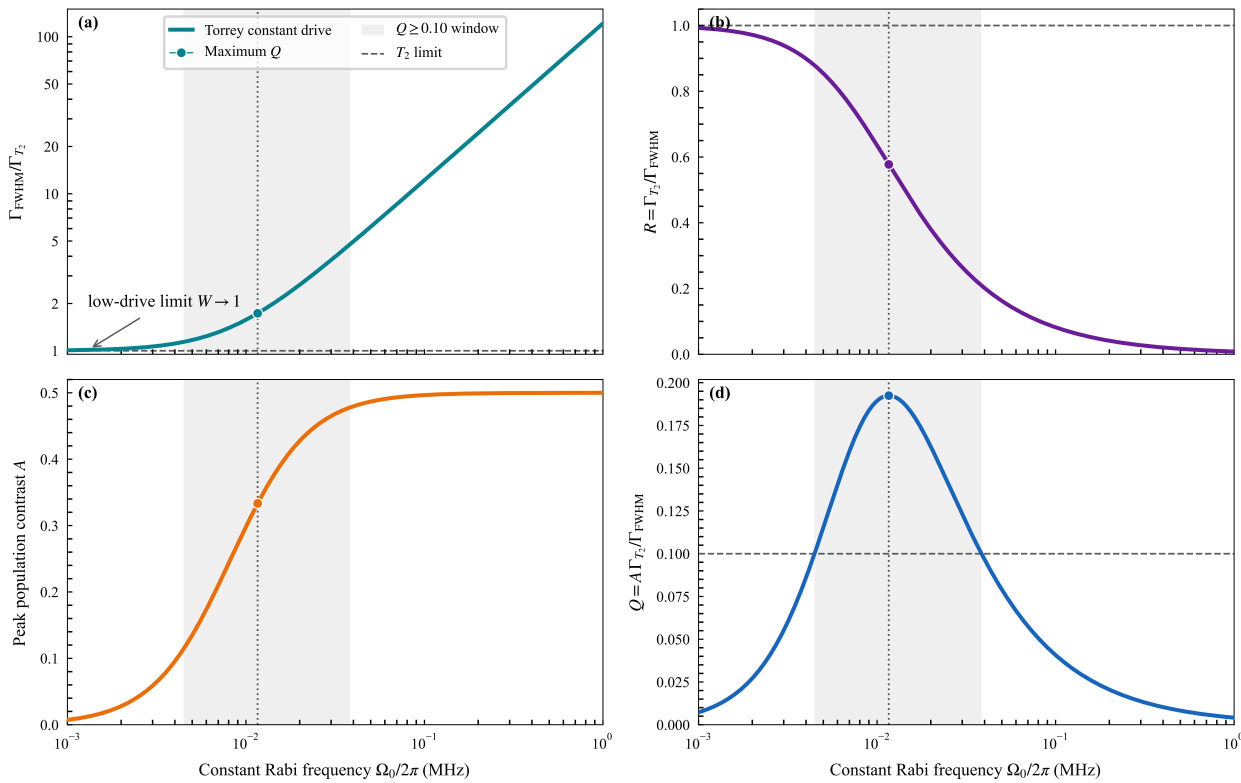

Figure S10. Analytic steady-state Torrey metrics for a constant drive. (a) FWHM in units of Γ_{T_2}; the curve explicitly approaches $W = 1$ in the low-drive limit. (b) Normalized resolution $R = \Gamma_{T_2}/\Gamma_{\mathrm{FWHM}}$. (c) Peak population contrast A. (d) Contrast-weighted resolution $Q = AR$. The shaded vertical band marks the interval $Q \geq 0.10$, and the dotted vertical line and circular markers identify the maximum of Q. Dashed horizontal lines indicate the T_2 reference in panels (a) and (b) and the operational Q threshold in panel (d). The constant-drive amplitude is shown on a logarithmic axis.

S9. ROOT-LORENTZIAN VERSUS ECHO-ROOT-LORENTZIAN PULSES

The two protocols are compared over the principal experimental control parameters: pulse duration, relative cutoff, and peak drive amplitude. For each parameter set, we examine the final spectroscopic response, extracted linewidth, signal contrast, and linewidth-to-signal ratio. Pulse-length sweeps distinguish the Fourier-limited and decoherence-limited regimes, while cutoff sweeps quantify broadening introduced by truncating the formally infinite Lorentzian tail. Drive-amplitude sweeps then test whether the narrow feature remains stable as the Rabi frequency is increased. These comparisons show where the phase inversion of the echo-root-Lorentzian pulse suppresses coherent oscillations and produces a broader, more uniform operating region than the corresponding root-Lorentzian pulse.

S9.1. Experimental Long-Pulse Comparison

At fixed cutoff and peak Rabi frequency, varying the pulse length separates short-pulse Fourier broadening from the long-pulse limit imposed by decoherence. The comparison also reveals amplitude-dependent coherent oscillations in the root-Lorentzian response and their suppression by the echo phase inversion.

The OPX1000 data archive contains a controlled pulse-duration series acquired on qubit q9 with matched root-Lorentzian and echo-root-Lorentzian scans spanning $L = 2$ to 250 μs. Figure S11 shows three representative durations across the series.

The complete measured series contains $L = 2$, 5, 10, 30, 60, 90, 130, 160, and 250 μs. All six scans displayed in Fig. S11 use the same -2.5 to $+2.5$ MHz detuning grid, amplitude-prefactor range, 100 repetitions per point, relative cutoff $c = 5 \times 10^{-4}$, and peak waveform amplitude. For $L = 160$ μs, the OPX1000 plays a 60 μs stored template stretched to the recorded physical duration. This preserves the normalized envelope while avoiding an unnecessarily long arbitrary waveform record.

The convergence at long duration has a simple physical interpretation. The pulse is long enough for the adiabatic passage to be completed for either phase convention, while T_1 relaxation during the drive washes out the protocol-dependent coherent population structure away from resonance. The broad off-resonant region is therefore driven toward the same saturated background for the root-Lorentzian and echo-root-Lorentzian pulses, leaving only an ultranarrow, resonance-centered feature. Once the Fourier width has fallen below the coherence scale, increasing L further cannot continue to narrow this feature; its width is expected to approach the T_2-limited value $2/T_2$ in angular-frequency units from Eq. (S8). The present q9 archive does not contain an independent T_2 characterization, so Fig. S11 demonstrates convergence and ultranarrowing but does not by itself establish a quantitative equality to the T_2 limit.

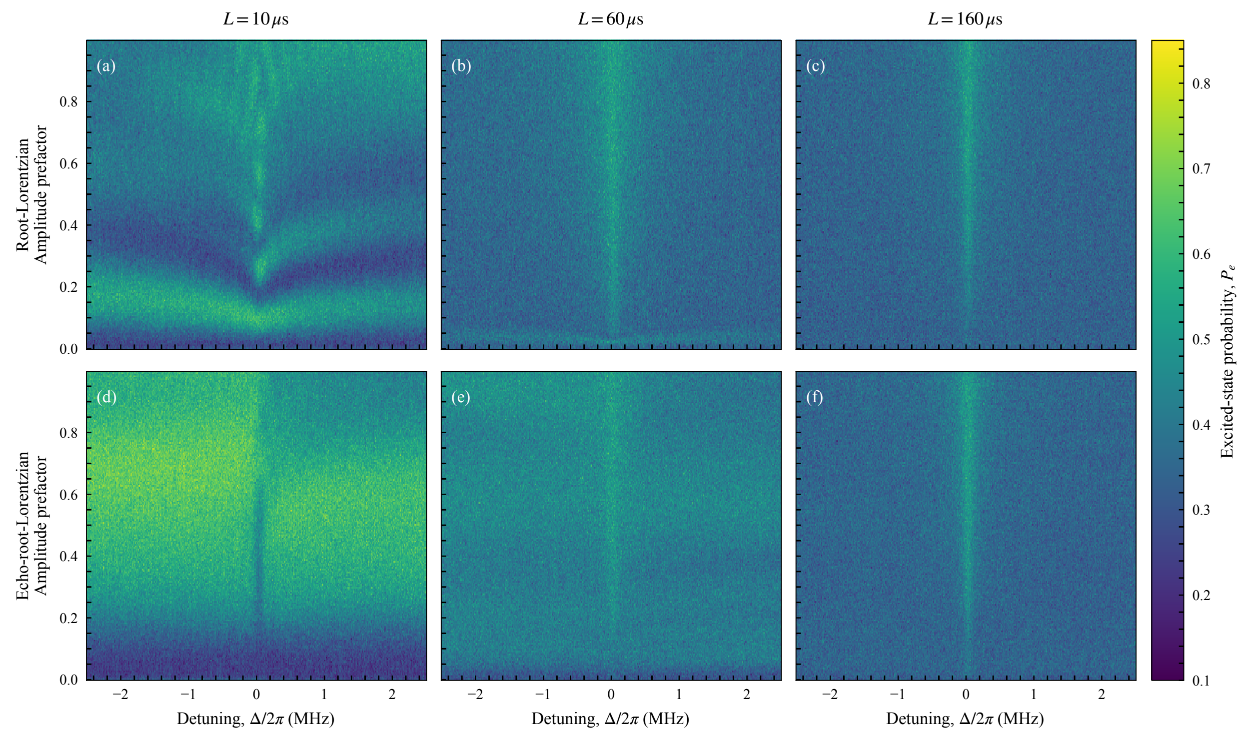

Figure S11. Experimental pulse-duration comparison on qubit q9. The top row shows root-Lorentzian spectroscopy and the bottom row shows the matched echo-root-Lorentzian protocol for $L = 10$, 60, and 160 μs. Each panel reports the measured excited-state probability over the same detuning and amplitude-prefactor grid; all panels share one color scale. The short root-Lorentzian scan exhibits pronounced amplitude-dependent structure, whereas the echo data are smoother. By $L = 160$ μs, both protocols have converged toward essentially the same response: a narrow resonance-centered line on a common relaxation-dominated background.

S9.2. Dependence on Pulse Cutoff

At fixed duration and peak Rabi frequency, varying the relative cutoff tests the sensitivity of both protocols to truncation of the Lorentzian tail. The relevant scale is the residual endpoint drive, $c\Omega_0$, which must remain small compared with the desired linewidth scale.

S9.3. Dependence on Drive Amplitude

Finally, amplitude sweeps compare the linewidth and contrast of the two pulse shapes over their full useful operating ranges. The echo protocol is assessed by the continuity and uniformity of its narrow-linewidth region, in addition to its minimum attainable linewidth.

S10. COMPARISON BETWEEN SIMULATION AND EXPERIMENT

Simulation and experiment are compared using the same final-state spectroscopic observable and the same linewidth-extraction procedure. The archived curves in the present figures use the historical coherence inputs described in Sec. S7, rather than the q1 values measured later on 2026-06-28. Their agreement with the data is therefore interpreted qualitatively as support for the echo-cancellation mechanism, not as a parameter-matched validation. A quantitative comparison requires regenerating the simulations with $T_1 = 51.24 \pm 0.73$ μs and either the conservative Ramsey reference $T_{2,\mathrm{eff}} = 7.31$ μs or a future measured Hahn-echo T_2. Systematic deviations at the largest amplitudes can also reflect physics omitted from the effective model, including higher-level coupling and strong-drive calibration effects.

S11. SIMULATED DURATION AND CUTOFF COMPARISON

Figures S12– S14 compare two-dimensional simulated spectroscopy maps for matched echo-root-Lorentzian and root-Lorentzian pulses. Each figure fixes the total duration at $L = 10$, 20, or 30 μs; the columns use relative endpoint cutoffs $c = 0.010$, 0.007, and 0.002. The upper row is the echo protocol and the lower row is the corresponding pulse without the central phase inversion. All panels use the same detuning, peak-Rabi-frequency, and color scales so that changes between protocols and cutoffs can be compared directly.

These preliminary maps are generated with the two-level Lindblad model of Sec. S7. They use the q1 measurement-linked values $T_1 = 51.24$ μs and $T_\phi = 7.87$ μs, corresponding to $T_{2,\mathrm{eff}} = 7.31$ μs. Here $T_{2,\mathrm{eff}}$ is the measured Ramsey T_2^* used as a conservative transverse time; it is not a measured Hahn-echo T_2. The maps span $\Delta/2\pi = -1$ to 1 MHz and $\Omega_0/2\pi = 0$ to 50 MHz. The generalized-Lorentzian exponent is $n = 1/2$, and the endpoint cutoff is imposed at fixed total duration using Eq. (S2). The nonuniform integration grid resolves the narrow pulse center and contains 1600 fourth-order Runge–Kutta steps per pulse half.

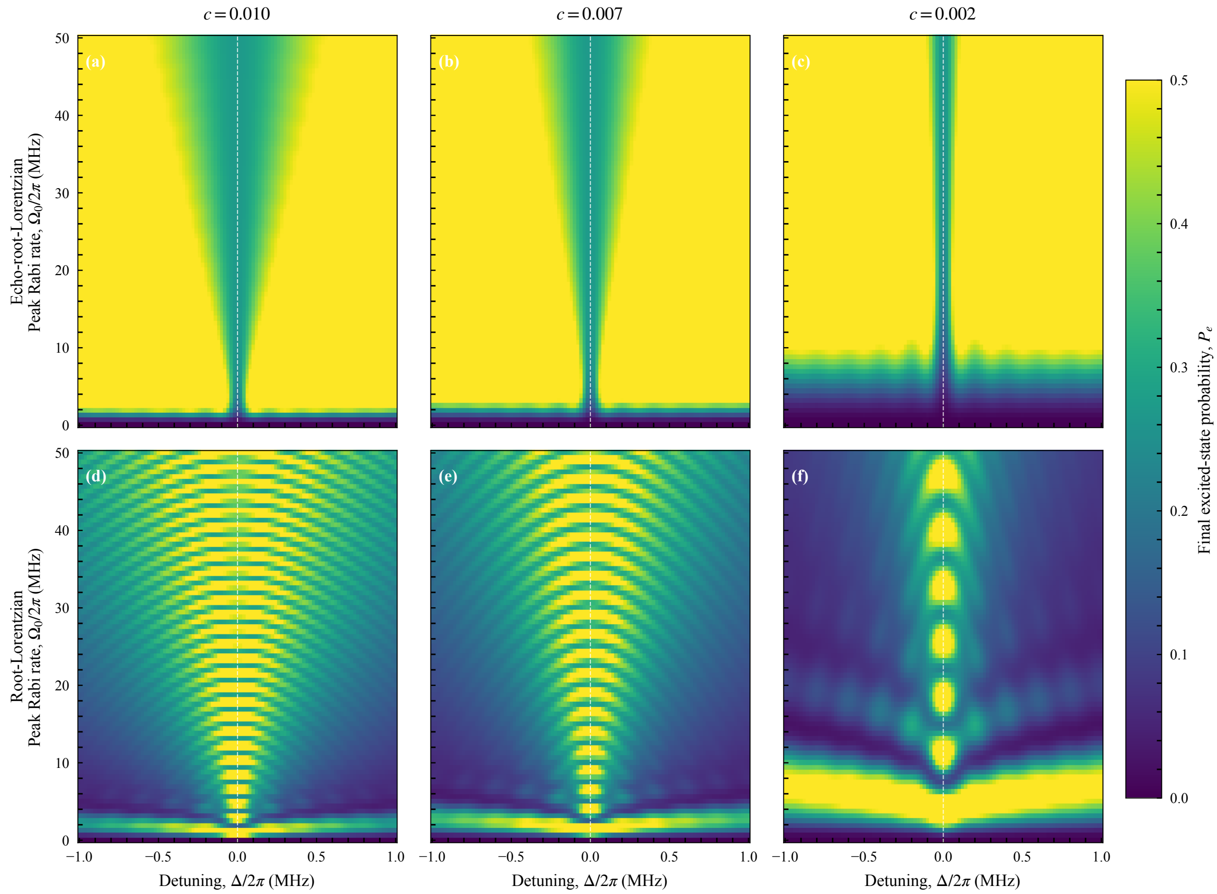

Figure S12. Simulated cutoff comparison for $L = 10\ \mu$s. The upper row shows echo-root-Lorentzian spectroscopy and the lower row shows the matched root-Lorentzian protocol. Columns correspond to $c = 0.010$, 0.007, and 0.002. Every panel displays final excited-state probability over the same detuning and peak-Rabi-frequency grid.

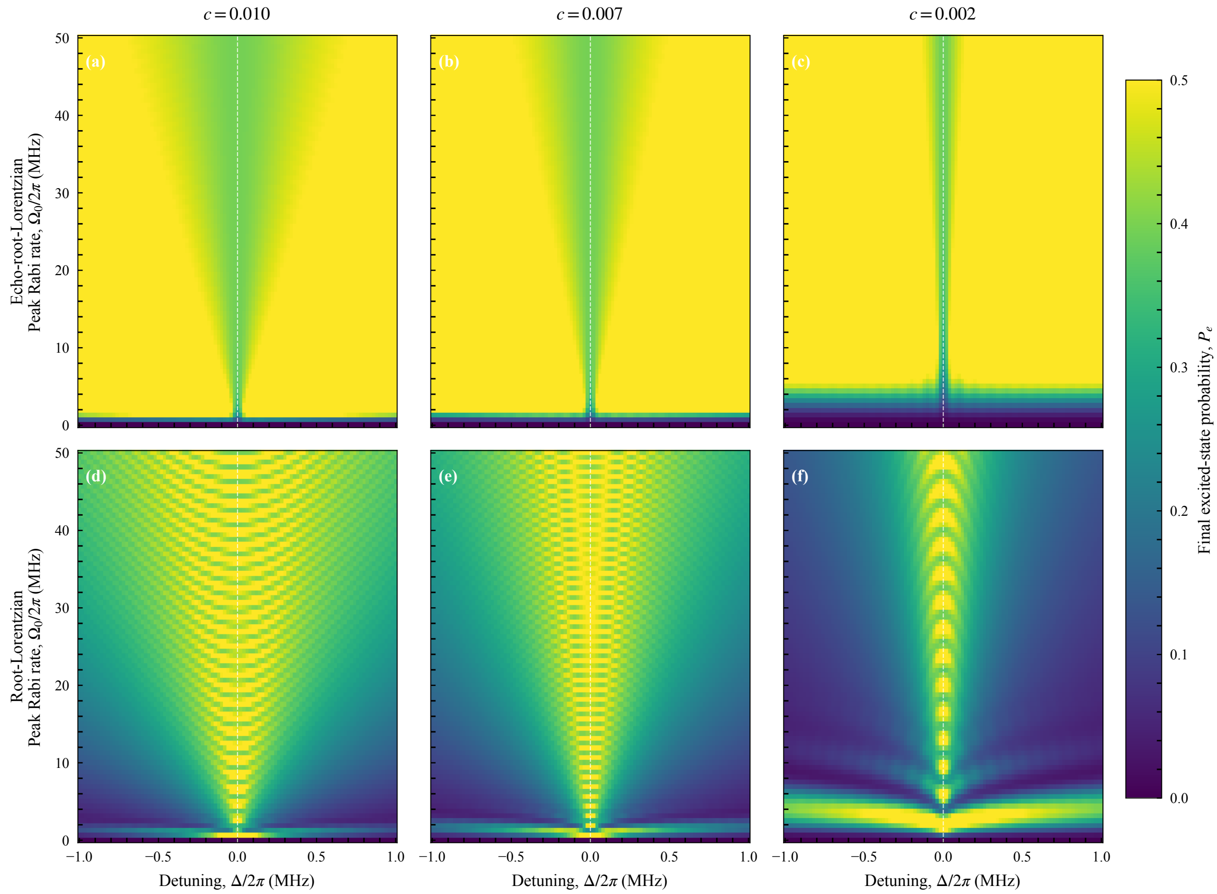

Figure S13. Simulated cutoff comparison for $L = 20$ μs, using the same layout, axes, color scale, and model parameters as Fig. S12.

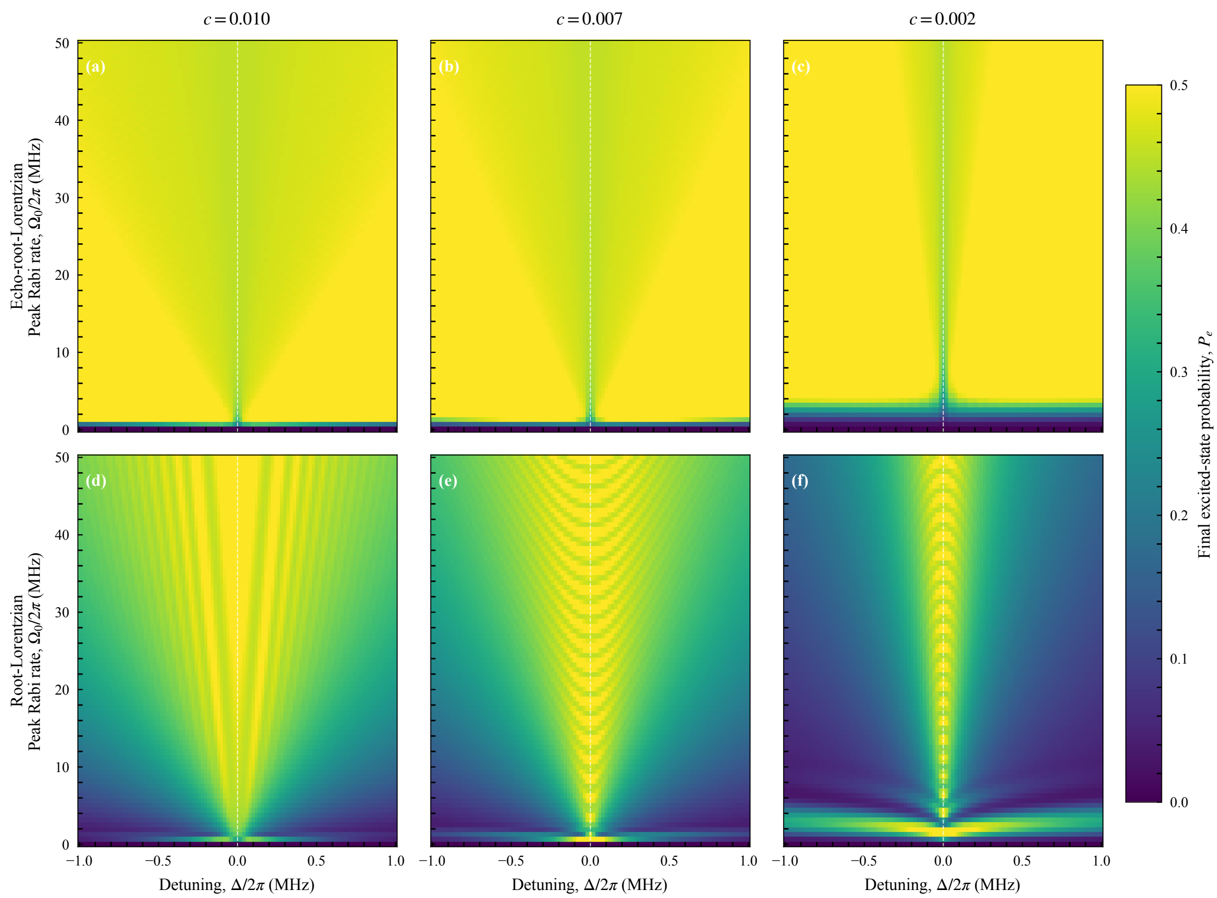

Figure S14. Simulated cutoff comparison for $L = 30$ μs, using the same layout, axes, color scale, and model parameters as Fig. S12.

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